

GRETN, MB, Canada

WMO: 714410

Lat: 49.0308N

Lon: 97.5600W

Elev: 829

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.7	-19.0	-30.5	1.0	-23.7	-25.7	1.4	-18.6	34.9	5.0	31.0	6.1	8.4	300	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	86.8	70.6	83.8	68.6	80.9	66.7	74.3	82.5	71.7	80.2	69.4	77.5	11.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	120.3	79.5	68.8	109.2	77.0	66.4	100.2	74.4	38.6	82.2	36.1	80.2	34.0	77.4	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.0	25.6	22.8	-29.7	93.2	6.3	2.4	-34.3	94.9	-38.0	96.3	-41.5	97.7	-46.1	99.5		
			-29.7	78.3	6.1	3.1	-34.0	80.5	-37.6	82.3	-41.0	84.0	-45.4	86.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	4.5	8.3	22.5	39.6	52.6	63.3	67.3	65.6	57.2	42.7	25.0	11.0
	DBStd	24.77	14.62	13.93	13.75	10.61	8.95	6.53	5.45	5.95	8.27	9.09	11.93	13.57
	HDD50	6099	1411	1169	854	335	77	2	0	0	26	264	752	1209
	HDD65	9978	1876	1589	1317	763	394	107	40	66	258	693	1201	1674
	CDD50	1877	0	0	1	22	158	400	535	483	241	37	0	0
	CDD65	282	0	0	0	0	10	55	110	83	23	1	0	0
	CDH74	3076	0	0	0	7	186	580	1044	943	297	19	0	0
	CDH80	799	0	0	0	0	48	139	255	283	71	3	0	0
	WSAvg	11.0	11.8	11.7	11.9	12.1	12.2	10.2	8.5	8.6	10.0	11.6	11.7	11.6
	PrecAvg	22.6	0.9	0.8	1.2	1.1	3.1	4.2	3.0	2.7	2.0	1.6	1.1	1.1
(15) Precipitation	PrecMax	28.1	1.8	2.0	3.4	2.1	5.9	9.4	7.0	6.3	4.6	4.5	3.1	1.9
	PrecMin	17.3	0.3	0.1	0.3	0.2	1.4	1.2	1.1	0.8	0.4	0.2	0.2	0.5
	PrecStd	3.1	0.4	0.4	0.8	0.6	1.2	2.0	1.6	1.7	1.0	1.2	0.8	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	41.2	60.3	74.2	86.0	89.8	89.2	91.5	86.4	76.3	57.6	42.1
		MCWB	35.2	38.3	50.3	55.1	62.4	70.8	75.1	71.4	67.7	60.1	49.5	36.5
	2%	DB	34.4	36.3	50.0	67.9	79.1	84.0	85.7	86.7	81.5	67.9	50.2	36.5
		MCWB	32.3	33.9	44.4	52.1	59.1	66.9	72.7	70.0	65.0	55.4	44.4	33.2
	5%	DB	30.4	33.1	43.4	62.8	74.2	80.5	82.8	82.9	77.2	62.3	44.7	32.6
		MCWB	28.9	31.7	39.5	49.9	56.6	64.9	70.2	68.3	62.9	52.9	40.7	31.0
	10%	DB	25.5	29.0	38.4	57.6	69.4	77.1	79.9	79.3	72.3	57.4	40.0	29.3
		MCWB	24.4	27.6	35.8	47.3	55.5	64.2	68.5	66.0	60.9	50.2	37.0	27.8
	0.4%	WB	35.7	38.7	52.5	58.0	66.7	76.3	79.1	76.6	72.1	62.0	50.3	36.9
		MCDB	38.3	41.7	58.4	68.7	77.0	83.6	86.1	85.7	81.9	71.2	55.2	41.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.5	34.2	44.5	53.9	62.7	71.4	75.1	72.7	68.1	57.5	44.6	33.5
		MCDB	34.4	35.9	49.4	64.1	73.6	79.0	82.4	82.5	76.2	65.8	49.3	35.8
	5%	WB	28.8	31.8	39.5	50.5	59.9	68.1	72.6	70.3	65.0	53.8	40.7	31.0
		MCDB	30.3	33.3	43.4	60.6	68.6	76.5	80.4	79.2	73.3	60.7	44.0	32.8
	10%	WB	24.4	27.6	35.9	47.7	57.3	65.6	70.2	68.0	62.2	50.3	37.0	28.0
		MCDB	25.5	29.0	38.0	56.9	66.7	74.0	78.0	76.8	70.5	57.3	39.7	29.3
	5% DB	MDBR	16.3	17.3	16.1	20.3	22.8	20.4	21.3	23.4	22.2	18.0	14.8	14.9
		MCDBR	19.0	17.0	20.1	30.7	30.5	25.9	24.5	28.2	29.2	26.8	21.5	18.0
		MCWBR	17.4	14.6	14.9	17.9	15.6	13.2	13.9	14.1	15.7	16.5	15.8	15.6
		MCDBR	18.9	15.8	19.1	27.6	23.5	21.0	22.6	24.3	24.9	24.6	20.0	17.4
(38) Clear-Sky Solar Irradiance	taub		0.274	0.299	0.305	0.355	0.371	0.382	0.386	0.382	0.347	0.306	0.274	0.255
		taud	2.271	2.215	2.392	2.348	2.362	2.358	2.354	2.359	2.451	2.538	2.492	2.393
	Ebn at Noon		248	267	289	283	282	279	275	271	271	268	253	245
		Edh at Noon	23	31	31	36	37	37	37	36	29	23	18	18
(44) All-Sky Solar Radiation	RadAvg		468	859	1248	1564	1739	1868	1989	1637	1196	723	411	324
	RadStd		49	62	114	132	127	142	83	106	85	100	55	37

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-2.21	-2.84	N/A	N/A	+124	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page